A compact RF power inverter with reduced EMI for a CubeSat electrothermal micro-thruster

Wei Liang[1], Xiaofan Cui[2], Luke Raymond[1], Lei Gu[1], Christine Charles[3], Rod Boswell[3], Juan Rivas-Davila[1]
[1]Stanford University, Stanford, CA 94305, USA
[2]University of Michigan, Ann Arbor, MI, 48105, USA
[3]The Australian National University, Canberra, ACT 2601, Australia

***Abstract*—This paper presents a compact RF power inverter implemented with a PCB multi-resonant input filter for driving an electrothermal plasma micro-thruster, mini Pocket Rocket 3.0 (miniPR 3.0). This miniPR 3.0 and its driving electronics are part of an investigation seeking to add micro propulsion systems to low cost CubeSats. The proposed RF power inverter and multi-resonant input filter use only air core inductors. All inductors are toroidal and formed by PCB traces and vias allowing a planar and compact design [1], [2]. The PCB structure serves as structural support to save space and volume for propulsion propellant and other payloads. Advantages of air core inductors include not being subject to saturation or variations from changing material properties of a magnetic core, as well as no Curie temperature rating. With the previous development [3] of successfully driving the miniPR 3.0 with a compact PCB RF power inverter, this paper emphasizes on addressing some of the EMI concerns. A lumped transmission line multi-resonant input filter is proposed to mitigate conductive EMI. Specifically, we present an 8th order multi-resonant input filter design and demonstrate the converter and the filter successfully driving the miniPR 3.0.**

I. Introduction

CubeSats, a sub-class of micro-satellites, are made up of standardized units of one or more 10 cm cubes. Recent popularity of the CubeSat platform is in part due to its low operation and launch costs that put "access to space" within the reach of academic institutions, small companies, and even individuals. There has also been interest in developing commercial applications using CubeSats including imaging/monitoring of resources, communication, asteroid mining, etc. CubeSats are sometimes released into orbit from the International Space Station (ISS) [4] and can be deployed from rockets as secondary payloads as the rocket ascends for its primary mission. Cubesats deployed from the ISS have an average lifetime of 6 to 12 months as they orbit in a decaying Low Earth Orbit (LEO) affected by atmospheric drag. Other CubeSats deployed from higher altitude rockets can last in space for years, but are often placed in non-circular or highly elliptic orbits that are sub-optimal for imaging tasks. Hence there is an increasing need for CubeSats to have micro-thruster systems for station keeping or orbital maneuvering.

Electric propulsion is a promising solution that has proven effective on large satellites and if miniaturized can be a significant asset to CubeSats. However, to date electric propulsion is not a sub-system commonly available in CubeSats due to difficulties associated with scaling thruster systems down to a suitable size. Different types of electric propulsion systems used in large satellites including Hall thrusters, RF ion thrusters, Helicon Double Layer thrusters, and various electrothermal plasma thrusters are currently being evaluated for CubeSats. Adequate power electronics are a significant barrier for the practical deployment of these systems, considering the large size and weight of conventional power supplies.

Previous work [3] presented a 13.8 MHz, 14 V input, 15 W RF power supply design that utilizes air core inductors to drive an electrothermal plasma micro-thruster (miniPR 3.0) intended for CubeSats. All the air core inductors in the power inverter design are toroidal and embedded in PCB resulting in a planar and compact design [1], [2] conducive to structrally supports for CubeSats to save space and volume for propulsion propellant and other payloads. Such inductors can operate in harsh environment due to the absence of magnetic cores. Moreover, the PCB inductors are implemented in the way that axial flux from the toroidal loop are canceled as in [2] allowing the whole power converter to be shielded with ground planes.

While adding Faraday shielding can easily take care of the radiative EMI, conductive EMI still needs to be addressed. In this paper, we proposed a multi-resonant input filter implemented with PCB inductors aimed at reducing conductive EMI. The filter is designed and verified to attenuate input current ripple at fundamental and a few harmonics of switching frequency. Moreover, we demonstrate a 13.9 MHz, 12 V input, 30 W class E inverter with the proposed multi-resonant filter successfully driving the electrothermal micro-thruster.

II. Multi-resonant PCB input filter design and verification

The proposed multi-resonant input filter is based on the synthesis of lumped transmission line analogs and largely outlined and presented by [5]. While Phinney, et al in [5] designed and implemented the lumped transmission line multi-resonant network for the purpose of waveform shaping MOSFET drain to source voltage in a class E inverter, the characteristics of this kind of multi-resonant network – particularly the ability to place impedance poles/zeros (maxima/minima) at arbitrary frequencies– is also attractive for input filter design. To design the network to serve input filtering functions, one solution is to align the maxima of the network input impedance (as shown in Fig. 1) at the fundamental and harmonics of the switching

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frequency (leaving the minima at halves between harmonics) and then connect the network in series at the converter input. Therefore, the high impedances at different harmonic frequencies can attenuate the current ripple. Fig. 2 gives an example of replacing the input choke with the proposed filter network for a Class E inverter, which also can reduce the energy storage by using resonant inductors instead of a choke inductor. There are many circuit networks that can synthesize the impedance plot in Fig. 1 with minimal component count. We choose the Cauer network implementation shown in Fig. 3 for its potential of integrating the components (L and C) with toroidal PCB inductors (e.g. illustrated in Fig. 4). Detailed on the synthesis of the circuit network process can be found in [6].

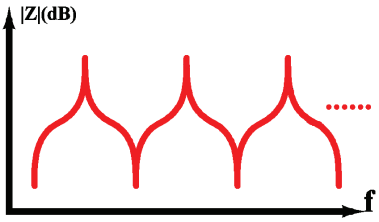


Fig. 1: transmission line impedance

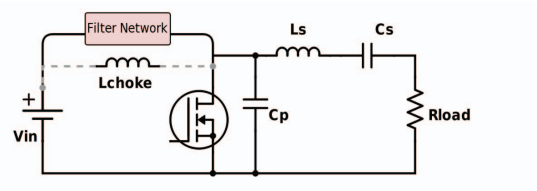


Fig. 2: replacing the input choke with filter network to reduce input current ripple

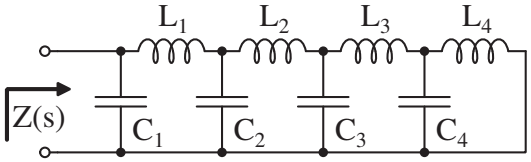


Fig. 3: 8 order Cauer network

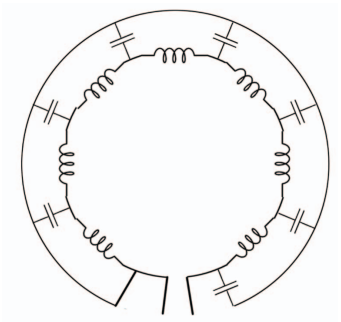


Fig. 4: Cauer network in a toroid shape

TABLE I: filter parameters

Parameter	Value	Units
f_s	27.12	MHz
L_1	168	nH
L_2	84	nH
L_3	84	nH
L_4	42	nH
C_1	21	pF
C_2	71	nF
C_3	110	pF
C_4	300	pF

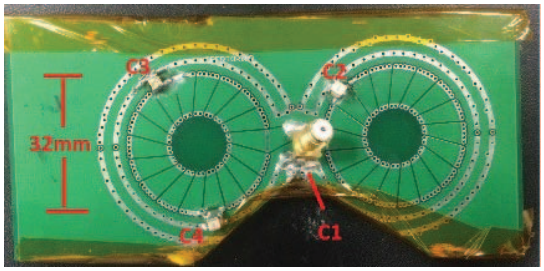


Fig. 5: PCB multi-resonant filter network prototype

For initial proof of concept, we designed and implemented an 8th order multi-resonant Cauer network that has impedance maxima located at the fundamental, 2nd, 3rd and 4th harmonic of the switching frequency. Design equations for inductances and capacitances are as following. A reasonable C_1 value needs to be determined as a starting point so that all values are comparable with other converter components.

$$L_1 = \frac{0.0986}{\omega^2 C_1} \quad (1)$$
$$C_2 = 3.174 C_1 \quad (2)$$
$$L_2 = \frac{0.0605}{\omega^2 C_1} \quad (3)$$
$$C_3 = 4.993 C_1 \quad (4)$$
$$L_3 = \frac{0.0513}{\omega^2 C_1} \quad (5)$$
$$C_4 = 13.42 C_1 \quad (6)$$
$$L_4 = \frac{0.0267}{\omega^2 C_1} \quad (7)$$

The component values are listed in Table I. Four inductors (L_1, L_2, L_3, L_4) are implemented within two series connected toroidal inductors printed on a 3.2mm thickness PCB, while four capacitors (C_1, C_2, C_3, C_4) are soldered at specific turns as shown in Fig. 5 to form the Cauer network as shown in Fig. 4. The input impedance of the prototype is measured and compared to the simulation as shown in Fig. 6. It can be seen that the measured impedance has close agreement with the simulation and thus it is expected to achieve good filtering result.

To verify the effectiveness of the filter network, we implement a 27.12 MHz, 100 V, 200 W class E RF inverter (parameter: MOSFET=ARF521; L_{choke}=1.5μH; L_s=260nH; C_s=254pF; C_p=9pF; R_{load}=12.5Ω; η=86%). The intend is to use the filter network in Fig. 5 to replace the commonly

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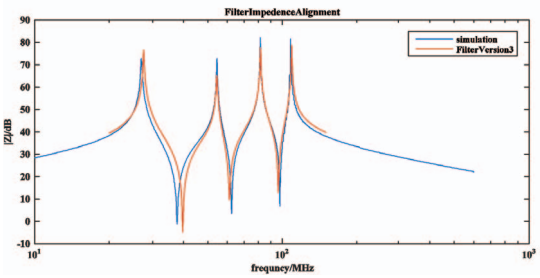


Fig. 6: measured vs simulated impedance

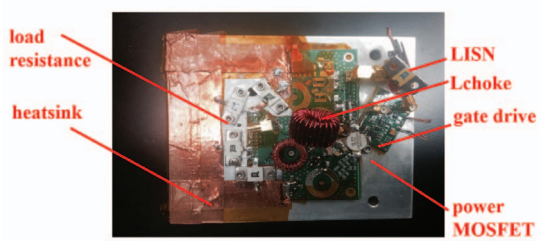


Fig. 7: 200W Class E inverter

designed input choke inductor and compare their input current ripple. Notice that in a normal class E inverter, the choke inductor can be chosen as arbitrarily large to achieve minimal input current ripple. However, the large choke inductor is not conducive to a compact RF inverter design. Therefore, in this case, the choke inductor is set as 1.5μH, which is close to the limit of an air core inductor can achieve in a reasonable form factor for targeted space related applications. The designed class E prototype is shown in Fig. 7. To measure input current ripples, a customized LISN (line impedance stabilization network) with extended frequency range is implemented and utilized. The LISN measurements of input current ripples of the Class E inverter with input choke and filter network are plotted in Fig. 8. It is seen that the filter network reduces the ripple significantly without any optimization on component quality factors or fine tuning of the filter network. Also note that the LISN is neither a standard copy nor connected to a spectrum analyzer/oscilloscope's 50 Ω input, thus the comparison of the two measurements is more important than the absolute voltage values.

To demonstrate that the use of the proposed filter network can reduce the energy storage requirement, start up transient is also studied for both cases (filter network vs. input choke) as shown in Fig. 9. Note that the waveform with filter case (less than 200 ns) reaches steady state faster than that with the input choke case (about 300 ns). Similar drain waveform shapes indicate that the filter network does not change the Class E inverter tuning significantly. The resulted filter network is also conducive to a compact design due to the reduction in overall input inductance, 378nH of the input filter network versus the 1.5μH input choke. In addition, applying the filter network leads to reduced requirement of input capacitance for the inverter, eliminating the use of low reliability capacitors particularly for space related applications.

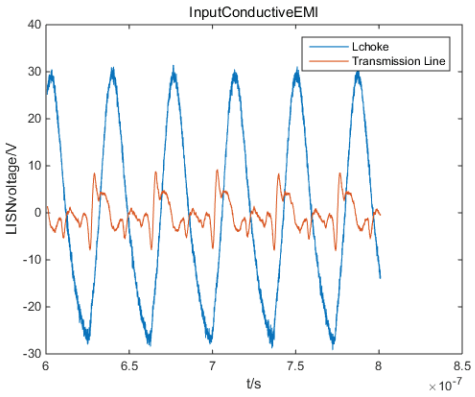


Fig. 8: LISN measurement of input ripple

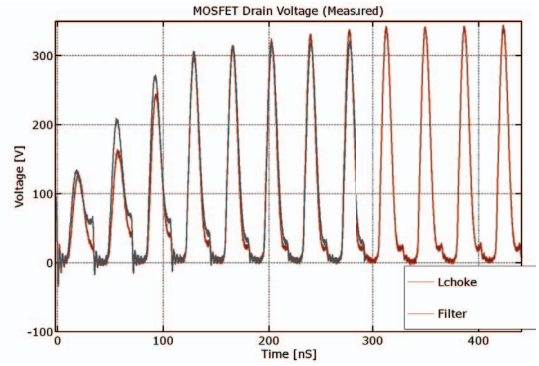


Fig. 9: measured MOSFET drain voltage comparison at start up transient

III. Driving a Plasma Load and the Electrothermal Micro-Thruster

With the proven effectiveness of the input filter, we designed a 13.9 MHz, 12 V input, 30 W class E inverter with the proposed filter network to drive the miniPR 3.0 (parameter: MOSFET=IPD530N15N3; L_s=180 nH; C_p=5 pF; C_s=2.11 nF) as shown in Fig. 10. The filter network is implemented by two series connected and vertically stacked toroidal PCB inductors on two 3.2 mm thickness PCBs for this particular prototype. The impedance plot is shown in Fig. 11. And the component values in the filter are listed in Table II.

TABLE II: filter parameters

Parameter	Value	Units
f_s	13.9	MHz
L_1	355	nH
L_2	218	nH
L_3	185	nH
L_4	96	nH
C_1	38	pF
C_2	120	nF
C_3	190	pF
C_4	510	pF

An additional L shaped high pass matching network (L_m=850 nH, C_m=115 pF) necessary to drive the plasma load and miniPR 3.0 (as shown in Fig. 12) is also implemented using two vertically stacked 3.2 mm thickness PCBs.

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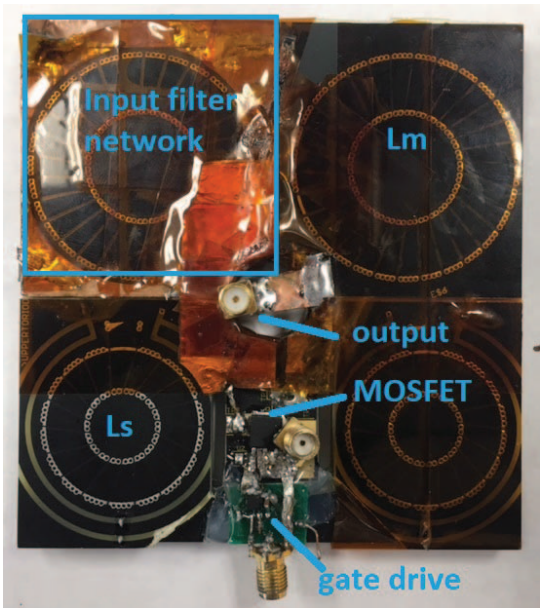


Fig. 10: 30W Class E inverter with input filter network and matching network for driving plasma loads. Dimensions of the board are 10 cm by 10 cm by 0.7 cm.

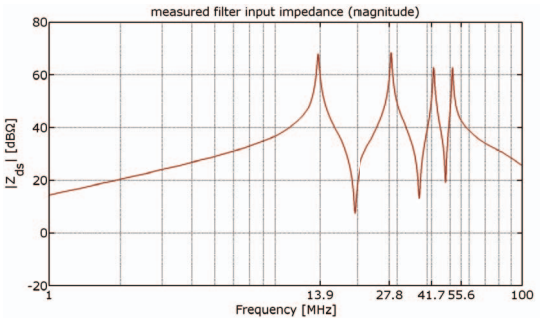


Fig. 11: measured filter network input impedance plot

Moreover, the whole inverter is formed by two 3.2 mm thickness PCBs and thus it has a planar shape with with a dimension of 10 cm by 10 cm by 0.7 cm. The physical dimension and mechanical strength (if using advanced PCB material/manufacturing techniques) of the prototype inverter make it suitable to serve as a structural support and/or a side panel for CubeSates to save precious space and volume for CubeSat propulsion propellant and other payloads.

First, to demonstrate the inverter can drive plasma loads, a quartz ampoule filled with low pressure Ar gas is used. Two electrodes are implemented on the Ar ampoule using copper tape as shown in Fig. 13. A capacitively coupled plasma (CCP) is expected to form inside the Ar ampoule when supplying RF power to this electrode configuration. Fig. 14 shows the inverter successfully strikes and runs the plasma in the Ar ampoule. Measured MOSFET drain voltage waveform is plotted in Fig. 15. The inverter was able to run the full 30 W power without requiring any additional heatsinking. To characterize the input current ripple, we also connect a customized LISN with extended frequency range. Fig. 16 shows the measured LISN voltage and it is clear that the fundamental component which contributes majority of conductive EMI has been minimized, compared to a normal class E inverter with input choke in which the fundamental component dominates the current ripple (such as the "L_{choke}" in Fig. 8). It is also worth to mention that the filter network significantly reduces the requirement of input capacitance to the extent that the inverter can drive the plasma normally without any input capacitance (long input power supply cables included).

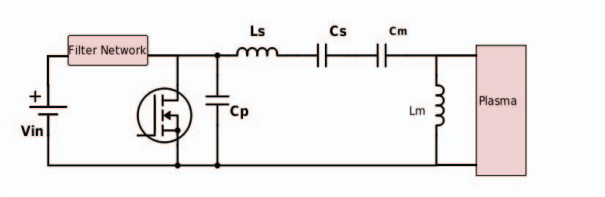


Fig. 12: schematic of the inverter running plasma load with a matching network (L_m and C_m)

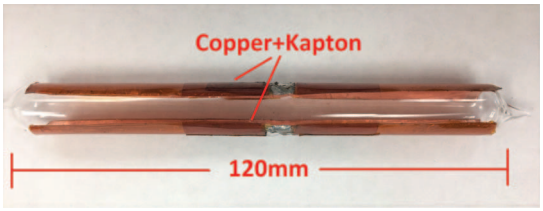


Fig. 13: low pressure Ar ampoule as a plasma load

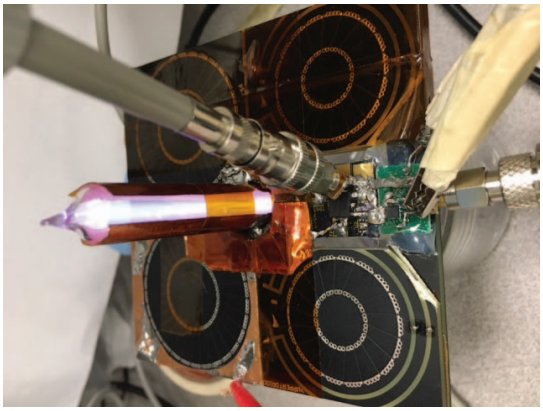


Fig. 14: inverter driving the plasma inside the Ar ampoule

The miniPR 3.0 consists of an 18-20 mm long, 1.5 mm inner diameter, 3 mm outer diameter alumina tube. A 5 mm wide, 16 mm diameter, annular copper powered electrode surrounds the alumina tube at the midpoint with the outer aluminum housing as the ground node, creating an asymmetric weakly ionized (less than 1%) capacitively coupled plasma (CCP) inside the tube (usually Ar) [7]. The schematic and assembly view of the miniPR 3.0 are shown in Fig. 17 and detailed description of the miniPR 3.0 can be found in [7]. The Pocket Rocket series electrothermal plasma micro-thrusters

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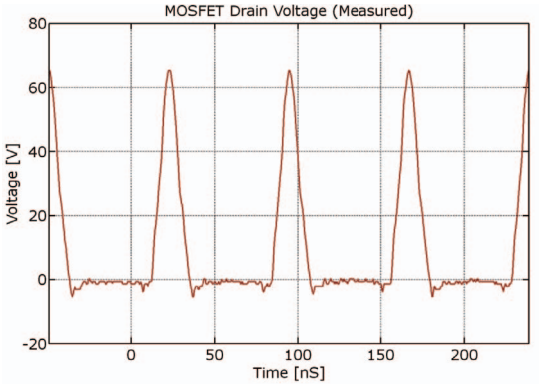


Fig. 15: Measured MOSFET drain voltage while driving the Ar ampoule

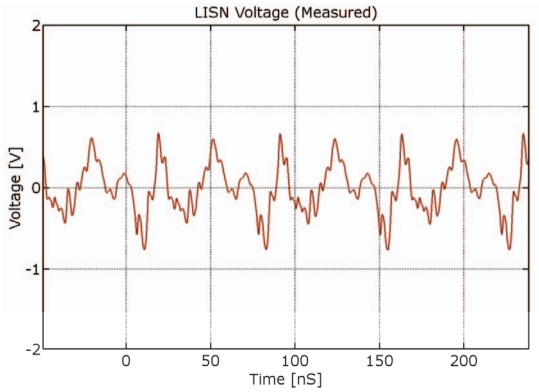


Fig. 16: Measured LISN voltage while driving the Ar ampoule

have been well modeled and studied in literature [8]–[11] although all previous development used bulky linear RF power amplifiers that prevents further development towards CubeSat form factors. The miniPR 3.0 setup is shown in Fig. 18 and it is the same as detailed in [3].

Fig. 19 shows the miniPR 3.0 plasma (Ar) running seen through the vacuum chamber viewport. The inverter is pulsed at 10% of the full 30 W power and the power can be controlled

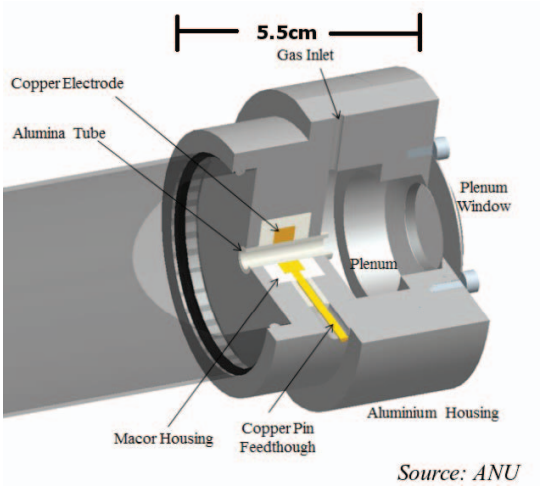


Fig. 17: inside the miniPR 3.0

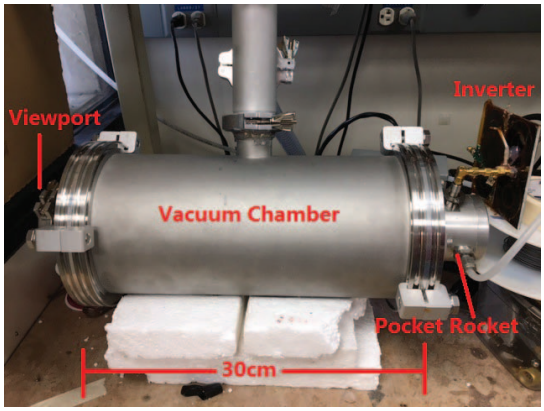


Fig. 18: vacuum chamber setup

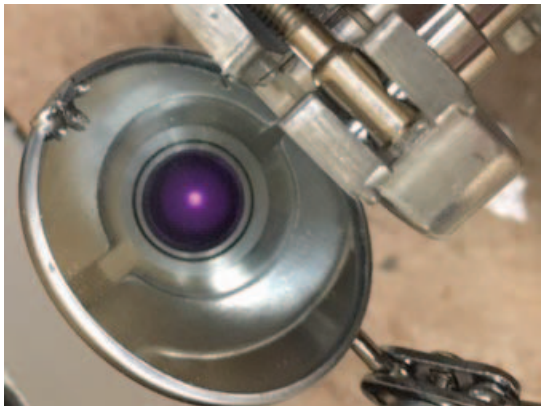


Fig. 19: Plasma running seen from the viewport using a surgeon mirror

by varying the pulsing duty. As our prototype vacuum chamber is located with the viewport facing against wall, a surgeon mirror is used to monitor the plasma through the viewport. Note that the Faraday shield is currently not implemented for the purpose of better demonstration.

IV. Conclusion

This paper presents a compact RF power inverter implemented with PCB multi-resonant input filter and later the Faraday shield for driving an electrothermal plasma microthruster, mini Pocket Rocket 3.0 (miniPR 3.0). This miniPR 3.0 and its driving electronics are low volume and are part of an investigation seeking to add propulsion to low cost CubeSats. Specifically, we present an 8th order multi-resonant input filter design and demonstrate the converter and the filter successfully driving the miniPR.

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